

C: Into the Onion

Source file name: onion.c, onion.cpp, onion.java, or onion.py

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A group of biologists is studying the structure of an onion. They cut the onion through its center and obtain a two-dimensional cross-section. In the resulting image, they identify N cells. For each cell, they record its position (x_i, y_i) and a measured value w_i (that's not worth explaining here).

The biologists observe that the cells form several layers, from the outside towards the center of the onion. The outermost layer L_1 , consists of all the cells lying on the boundary of the smallest convex polygon that contains all N cells, including not only its vertices, but also any cell that lies exactly on one of its edges. After removing those cells of L_1 , layer L_2 is determined the same way from remaining cells: it consists of all the remaining cells lying on the boundary of the smallest convex polygon that contains them. This process is repeated, layer after layer, until no cell remains outside the last convex polygon obtained, that is, until every remaining cell lies on its boundary. That last layer is called the *heart* of the onion.

The layers are numbered from the outside towards the center, so L_1 is the outermost layer and L_K is the heart of the onion (see Figure 1). Let s_i denote the number of cells in layer L_i .

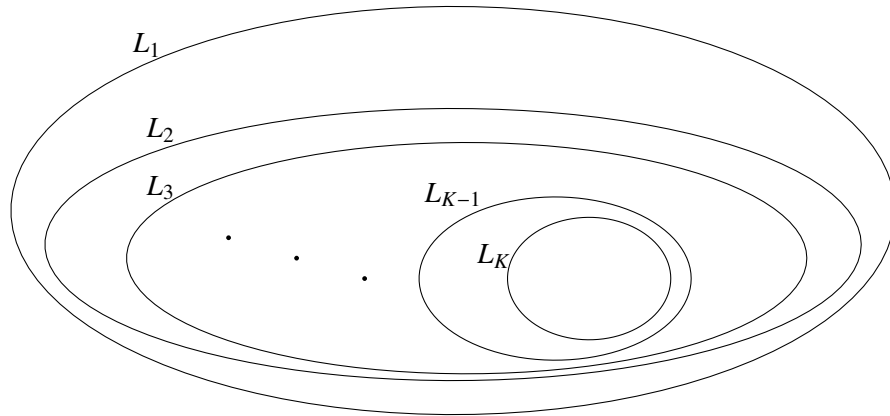


Figure 1: Example of a two-dimensional cross-section obtained by cutting an onion.

The density of a layer affects how easily a chemical marker can move through the onion. For a cell u in layer L_i , the biologists define the *transmission field* of u as a disk with center in u and radius $\frac{3}{2} \cdot d(L_i, L_{i+1})$ where $d(L_i, L_{i+1})$ is the minimum distance between the cells in the layer L_i and the cells in the layer L_{i+1} .

If v is a cell in the next layer L_{i+1} , the marker can pass from u to v exactly when v lies inside the transmission field of u , that is, when the distance between u and v is less than or equal to the radius of the transmission field of u . Note that the marker can only advance from a layer to the immediately next layer (see Figure 2).

The marker can be introduced into any cell of the outermost layer. From there, it repeatedly moves towards the center, always going from one layer to the next according to the rule above. Whenever the marker enters a cell, the biologists collect its measured value w_i (including the cell where the marker was initially introduced). The value of an experiment is the sum of the values collected along the entire sequence of cells visited by the marker. If there is more than one possible sequence, the biologists are interested only in the one with the maximum value.

For example, suppose 7 cells are identified (see Figure 3):

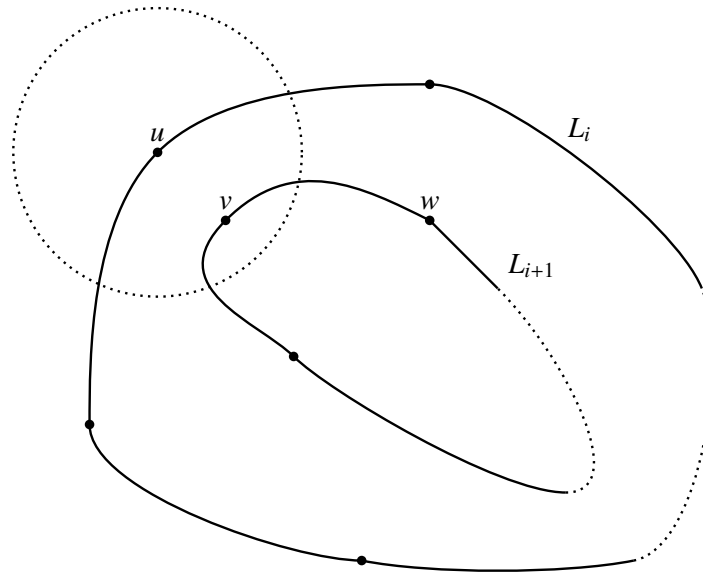


Figure 2: The cell v lies inside the transmission field of u , but the cell w lies out the transmission field of u . The transmission field of u is the circle with center in u and radius $\frac{3}{2} \cdot d(L_i, L_{i+1})$.

Cell number	x	y	w
1	2	2	1
2	-2	2	1
3	-2	-2	1
4	3	-2	1
5	2	0	5
6	0	1	3
7	-1	-1	8

The minimum distance between L_1 and L_2 is reached between cell 3 and the cell 7.

If the marker is introduced in c_1 , it can pass to c_5 . If the marker is introduced in c_3 , it can pass to c_7 . The best choice is to introduce the marker at c_3 : the value of the experiment is $w_3 + w_7 = 1 + 8 = 9$.

The team needs to determine the maximum value that can be obtained by introducing the marker at some cell of the outermost layer and successfully carrying it to one of the cells in the heart of the onion.

If no valid propagation reaches heart, the situation must be notified.

Input

There are several test cases.

Each test case starts with a line containing an integer N , the number of cells. Each of the next N lines contains three integers x_i , y_i and w_i describing a cell: (x_i, y_i) is the position in the xy -plane and w_i the measured value of the cell. No two cells share the same position. $1 \leq N \leq 3000$, $-10^8 \leq x_i, y_i \leq 10^8$ and $1 \leq w_i \leq 10^6$.

The input end is signaled with a line with $N = 0$ (that should not be processed).

The input must be read from standard input.

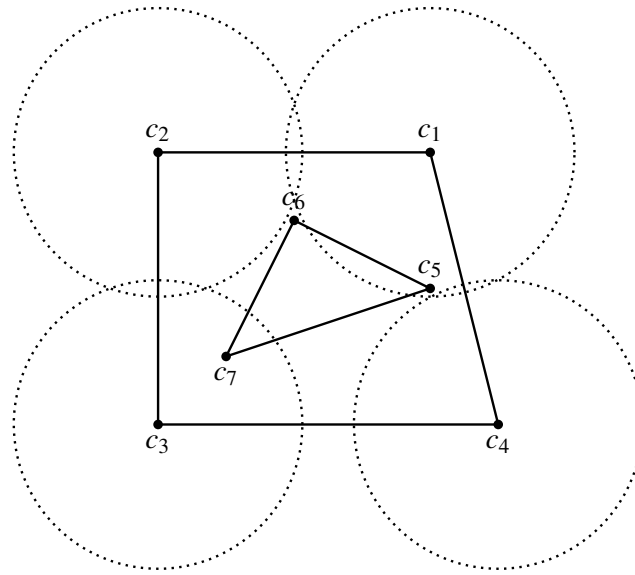


Figure 3: In the example the marker can be introduced into c_1 , c_2 , c_3 and c_4 . The marker can pass from c_1 to c_5 and from c_3 to c_7 .

Output

A single line per case: the maximum value that can be obtained by a valid propagation from the outermost layer to the heart of the onion. If no valid propagation reaches the heart, print -1.

The output must be written to standard output.

Sample Input	Sample Output
14 0 4 3 -4 3 1 -5 -1 1 -1 -3 3 3 -3 3 4 1 3 0 2 10 1 1 1 -3 2 2 2 -2 3 -2 0 3 0 1 100 0 0 30 -1 0 3 0	113